

1) (Murphy 2.1b) Beta Distribution

- given $\theta \sim \text{Beta}(a, b)$ with probability density

$$p(\theta; a, b) = \frac{1}{B(a, b)} \theta^{a-1} (1-\theta)^{b-1} \quad 0 \leq \theta \leq 1, \text{ where } B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

- deriv: $E[\theta]$, $\text{mode}(\theta)$, $\text{Var}(\theta)$

(a) Derive the mean:

$$E[\theta] = \int_0^1 \theta p(\theta) d\theta$$

$$E[\theta] = \int_0^1 \theta \frac{1}{B(a, b)} \theta^{a-1} (1-\theta)^{b-1} d\theta = \frac{1}{B(a, b)} \int_0^1 \theta^a (1-\theta)^{b-1} d\theta$$

$$E[\theta] = \frac{1}{B(a, b)} \int_0^1 \theta^a (1-\theta)^{b-1} d\theta$$

Recall Beta function definition:

$$B(r, s) = \int_0^1 t^{r-1} (1-t)^{s-1} dt \quad \text{where} \quad \int_0^1 \theta^a (1-\theta)^{b-1} d\theta$$

$$r-1 = a \quad \text{so} \quad r = a+1$$

therefore,

$$\int_0^1 \theta^a (1-\theta)^{b-1} d\theta = B(a+1, b)$$

$$\text{Thus, } E[\theta] = \frac{B(a+1, b)}{B(a, b)}$$

- we can still simplify using gamma functions

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \quad \text{so} \quad B(a+1, b) = \frac{\Gamma(a+1)\Gamma(b)}{\Gamma(a+1+b)}$$

$$E[\theta] = \frac{\frac{\Gamma(a+1)\Gamma(b)}{\Gamma(a+1+b)}}{\frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}} = \frac{\Gamma(a+1)\Gamma(b)}{\Gamma(a+1+b)} \cdot \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} = \frac{\Gamma(a+1)\Gamma(a+b)}{\Gamma(a+1+b)\Gamma(a)}$$

- gamma identity: $\Gamma(z+1) = z\Gamma(z)$

$$E[\theta] = \frac{a\Gamma(a)\Gamma(a+b)}{(a+b)\Gamma(a+b)\Gamma(a)}$$

$$E[\theta] = \frac{a}{a+b}$$

(b) derive the mode

- mode is the value of θ where the probability density is maximized

$$\text{argmax}_{\theta} \left[\frac{1}{B(a, b)} \theta^{a-1} (1-\theta)^{b-1} \right], \quad \frac{1}{B(a, b)} \text{ doesn't depend on } \theta \text{ so doesn't affect where maximum occurs}$$

- so only need to maximize: $\theta^{a-1} (1-\theta)^{b-1}$

~ take log to make deriving easier

$$\log[p(\theta)] = -\log B(a, b) + (a-1)\log \theta + (b-1)\log(1-\theta)$$

$$\frac{\partial}{\partial \theta} [-\log B(a, b)] = 0 \quad \frac{\partial}{\partial \theta} [(a-1)\log \theta + (b-1)\log(1-\theta)] = -\frac{b-1}{1-\theta}$$

$$\frac{\partial}{\partial \theta} [(a-1)\log \theta] = \frac{a-1}{\theta}$$

- mode means set derivative = 0

$$0 = 0 + \frac{a-1}{\theta} - \frac{b-1}{1-\theta}$$

$$\frac{a-1}{\theta} = \frac{b-1}{1-\theta}$$

$$a-1(1-\theta) = b-1(\theta)$$

$$(a-1) - \theta(a-1) = \theta(b-1)$$

$$a-1 = \theta(b-1) + \theta(a-1)$$

$$a-1 = \theta[(b-1) + (a-1)]$$

$$\theta_{\text{mode}} = \frac{(a-1)}{(b-1) + (a-1)}, \quad a > 1 \text{ and } b > 1$$

(c) derive variance θ

- variance identity: $\text{Var}(\theta) = E[\theta^2] - (E[\theta])^2$

$$E[\theta] = \frac{a}{a+b}$$

$$E[\theta^2] = \int_0^1 \theta^2 p(\theta) d\theta = \frac{1}{B(a, b)} \int_0^1 \theta^2 \theta^{a-1} (1-\theta)^{b-1} d\theta = \frac{1}{B(a, b)} \int_0^1 \theta^{a+1} (1-\theta)^{b-1} d\theta = B(a+2, b)$$

$$E[\theta^2] = \frac{B(a+2, b)}{B(a, b)}$$

- using gamma functions

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} \quad \text{so} \quad B(a+2, b) = \frac{\Gamma(a+2)\Gamma(b)}{\Gamma(a+b+2)}$$

$$E[\theta^2] = \frac{\Gamma(a+2)\Gamma(b)}{\Gamma(a+b+2)} \cdot \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} = \frac{\Gamma(a+2)\Gamma(a+b)}{\Gamma(a+b+2)\Gamma(a)}$$

- using gamma identities: $\Gamma(a+2) = (a+1)a\Gamma(a)$
 $\Gamma(a+b+2) = (a+b+1)(a+b)\Gamma(a+b)$

$$E[\theta^2] = \frac{(a+1)a\Gamma(a)\Gamma(a+b)}{(a+b+1)(a+b)\Gamma(a+b)\Gamma(a)} = \frac{(a+1)a}{(a+b+1)(a+b)}$$

$$\text{Var}(\theta) = E[\theta^2] - (E[\theta])^2$$

$$= \frac{a(a+1)}{(a+b+1)(a+b)} - \left(\frac{a}{a+b}\right)^2$$

$$= \frac{a(a+1)}{(a+b+1)(a+b)} - \frac{a^2}{(a+b)^2}$$

$$\left(\begin{aligned} \frac{a(a+1)}{(a+b+1)(a+b)} - \frac{a^2}{(a+b)^2} &= \frac{a(a+1)(a+b) - a^2(a+b+1)}{(a+b+1)(a+b)^2} \\ \frac{a^2}{(a+b)^2} &= \frac{a^2(a+b+1)}{(a+b+1)(a+b)^2} \end{aligned} \right)$$

$$= \frac{a(a+1)(a+b) - a^2(a+b+1)}{(a+b+1)(a+b)^2}$$

$$= \frac{a(a+1)(a+b) - a^2(a+b+1)}{(a+b+1)(a+b)^2}$$

$$= \frac{a[(a+1)(a+b) - a(a+b+1)]}{(a+b+1)(a+b)^2}$$

$$= \frac{a[a^2 + ab + ab + b - a^2 - ab - a]}{(a+b+1)(a+b)^2}$$

$$\text{Var}(\theta) = \frac{a(b)}{(a+b+1)(a+b)^2}$$

2) multinomial / categorical distribution

$$\text{Cat}(x | \mu) = \prod_{i=1}^K \mu_i^{x_i}$$

- show it belongs to the exponential family & the corresponding general linear model (GLM)

- is the same as multinomial logistic regression (softmax regression)

- recall multinomial distribution:

K discrete categories, outcome $x \rightarrow$ one-hot vector
 parameter: $\mu_i \rightarrow$ probability of category i , with $\sum_{i=1}^K \mu_i = 1$

(a) belonging to the exponential family

- recall a probability is in the exponential family if it can be expressed in the form:

$$p(x | \eta) = h(x) \exp[\eta^T T(x) - A(\eta)]$$

where: η = natural parameter, $T(x)$ = sufficient statistic, $A(\eta)$ = log partition, $h(x)$ = base measure

$$p(x | \mu) = \prod_{i=1}^K \mu_i^{x_i} \Rightarrow p(x | \eta) = h(x) \exp[\eta^T T(x) - A(\eta)]$$

- identity: $a^b = e^{b \log a}$ so, $\mu_i^{x_i} = e^{x_i \log \mu_i} = \exp(x_i \log \mu_i)$

$$p(x | \mu) = \prod_{i=1}^K \exp(x_i \log \mu_i)$$

- identity: $a^b a^c = a^{b+c}$ so, $p(x | \mu) = \exp\left(\sum_{i=1}^K x_i \log \mu_i\right)$

- constraint on μ_i : $\mu_1 + \mu_2 + \dots + \mu_K = 1$

$\hookrightarrow$ all K probabilities are not independent

$$\sum_{i=1}^K x_i \log \mu_i = \sum_{i=1}^{K-1} x_i \log \mu_i + x_K \log \mu_K$$

- because x is one-hot

$$x_K = 1 - \sum_{i=1}^{K-1} x_i$$

- substitute x_K :

$$= \sum_{i=1}^{K-1} x_i \log \mu_i + \left(1 - \sum_{i=1}^{K-1} x_i\right) \log \mu_K$$

$$= \sum_{i=1}^{K-1} x_i \log \mu_i + \left(\log \mu_K - \log \mu_K \left(\sum_{i=1}^{K-1} x_i\right)\right)$$

$$= \sum_{i=1}^{K-1} x_i (\log \mu_i - \log \mu_K) + \log \mu_K$$

$$= \sum_{i=1}^{K-1} x_i \log \left(\frac{\mu_i}{\mu_K}\right) + \log \mu_K$$

- define natural parameters

$$\eta_i = \log \left(\frac{\mu_i}{\mu_K}\right), \quad i = 1, \dots, K-1$$

- probability becomes:

$$p(x | \mu) = \exp\left[\sum_{i=1}^{K-1} x_i \eta_i + \log \mu_K\right]$$

Goal: $\exp[\eta^T T(x) - A(\eta)] \rightarrow$ rewrite $\log \mu_K$ in terms of η

$$\exp[\eta_i] = e^{\eta_i} = \frac{\mu_i}{\mu_K} \Rightarrow \mu_i = \mu_K e^{\eta_i}$$

$$\mu_K + \sum_{i=1}^{K-1} \mu_i = 1 \Rightarrow \mu_K + \sum_{i=1}^{K-1} \mu_K e^{\eta_i} = 1$$

$$\mu_K \left(1 + \sum_{i=1}^{K-1} e^{\eta_i}\right) = 1$$

$$\mu_K = \frac{1}{1 + \sum_{i=1}^{K-1} e^{\eta_i}} \Rightarrow \mu_i = \frac{e^{\eta_i}}{1 + \sum_{i=1}^{K-1} e^{\eta_i}}$$

- find $A(\eta)$

$$\log \mu_K = \log \left[\frac{1}{1 + \sum_{i=1}^{K-1} e^{\eta_i}} \right] \quad \text{identity: } \log\left(\frac{1}{a}\right) = -\log a$$

$$\log \mu_K = -\log \left[1 + \sum_{i=1}^{K-1} e^{\eta_i} \right]$$

$$p(x | \eta) = \exp\left[\sum_{i=1}^{K-1} x_i \eta_i - \log \left(1 + \sum_{i=1}^{K-1} e^{\eta_i}\right)\right]$$

$$\text{compare: } p(x | \eta) = h(x) \exp[\eta^T T(x) - A(\eta)]$$

$$\hookrightarrow T(x) = x_1, \dots, x_{K-1} \quad \text{and} \quad A(\eta) = \log \left(1 + \sum_{i=1}^{K-1} e^{\eta_i}\right) \quad \text{and} \quad h(x) = 1$$

Therefore, $\text{Cat}(x | \mu)$ is in the exponential family

(b) show the GLM gives softmax regression

- recall, in GLM, we model the natural parameter η_i as a linear function of the input features

let the input vector be z . Then for category i :

$$\eta_i = w_i^T z, \quad \text{for } i = 1, \dots, K-1$$

$$\mu_i = \frac{e^{\eta_i}}{1 + \sum_{j=1}^{K-1} e^{\eta_j}} \quad \text{substitute } \eta_i = w_i^T z : \mu_i = \frac{e^{w_i^T z}}{1 + \sum_{j=1}^{K-1} e^{w_j^T z}}, \quad \text{for } i = 1, \dots, K-1$$

$$\mu_K = \frac{1}{1 + \sum_{j=1}^{K-1} e^{w_j^T z}}$$

- introducing softmax

$$P(Y = i | z) = \frac{e^{w_i^T z}}{\sum_{j=1}^K e^{w_j^T z}}$$

- recall category K is our reference category so we can set its parameter to zero:

$$w_K = 0 \Rightarrow w_K^T z = 0 \Rightarrow e^{w_K^T z} = e^0 = 1$$

- so,

$$1 + \sum_{j=1}^{K-1} e^{w_j^T z} = \sum_{j=1}^K e^{w_j^T z}$$

- therefore,

$$P(Y = i | z) = \frac{\exp(w_i^T z)}{\sum_{j=1}^K \exp(w_j^T z)} \quad \text{which is exactly the softmax function}$$